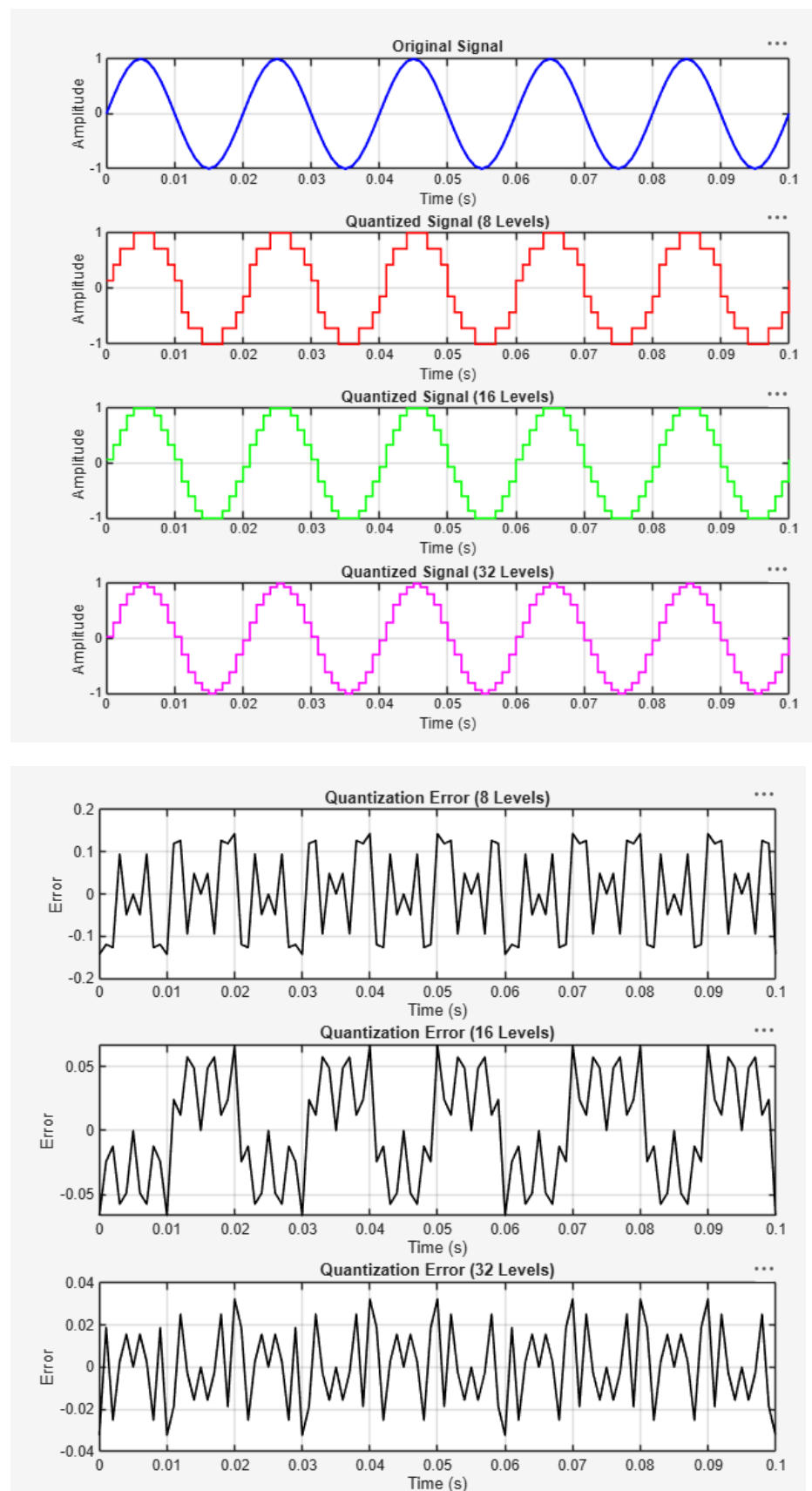


6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

```
1  clc;
2  clear;
3  close all;
4
5  %% Signal
6  fs = 1000;
7  f = 50;
8  t = 0:1/fs:0.1;
9  x = sin(2*pi*f*t);
10
11 %% Quantization Levels
12 L = [8 16 32];
13 colors = {'r','g','m'};
14
15 figure;
16
17 for i = 1:length(L)
18
19     % Uniform Quantization
20     xq{i} = round((x + 1) * (L(i) - 1) / 2) * (2 / (L(i) - 1)) - 1;
21
22     % Error & MSE
23     e{i} = x - xq{i};
24     mse(i) = mean(e{i}.^2);
25
26     % Plot Quantized Signals
27     subplot(4,1,i+1);
28     stairs(t, xq{i}, colors{i}, 'LineWidth', 1.2);
29     title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
30     xlabel('Time (s)');
31     ylabel('Amplitude');
32     grid on;
33
34 end
35
36 %% Original Signal
37 subplot(4,1,1);
38 plot(t, x, 'b', 'LineWidth', 1.5);
39 title('Original Signal');
40 xlabel('Time (s)');
41 ylabel('Amplitude');
42 grid on;
43
44 %% Display MSE
45 fprintf('MSE for 8 Levels = %f\n', mse(1));
46 fprintf('MSE for 16 Levels = %f\n', mse(2));
47 fprintf('MSE for 32 Levels = %f\n', mse(3));
48
49 %% Error Plots
50 figure;
51
52 for i = 1:3
53     subplot(3,1,i);
54     plot(t, e{i}, 'k', 'LineWidth', 1.2);
55     title(['Quantization Error (' num2str(L(i)) ' Levels)']);
56     xlabel('Time (s)');
57     ylabel('Error');
58     grid on;
59 end
```

Output:



6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation Rounding Techniques in MATLAB

```
1  clc; clear; close all
2  n=0:79;
3  x=sin(2*pi*n/40);
4  q=8;
5  xt=floor(x*q)/q;
6  xr=round(x*q)/q;
7  et=x-xt;
8  er=x-xr;
9  % Fig 1
10 figure
11 plot(n,x,'b',n,xr,'r','LineWidth',1.5)
12 grid on
13 title('Original vs Quantized')
14 legend('Original','Quantized')
15 % Fig 2
16 figure
17 plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)
18 grid on
19 title('Truncation vs Rounding')
20 legend('Original','Truncated','Rounded')
21 % Fig 3
22 figure
23 stem(n,et,'r')
24 hold on
25 stem(n,er,'g')
26 grid on
27 title('Quantization Error')
28 legend('Truncation','Rounding')
29 % Fig 4
30 figure
31 bar([mean(et.^2) mean(er.^2)])
32 set(gca,'XTickLabel',{'Truncation','Rounding'})
33 grid on
34 title('MSE Comparison')
35 ylabel('MSE')
```

Output:

